

Relatório – Exercício Prático 6 – Sophia Carrazza

Parte 1 – Respostas

1. O que é um arquivo fonte?

Resp: Letra A - um arquivo de texto que contém instruções de linguagem de programação.

2. O que é um registrador?

Resp: Letra B - uma parte do processador que possui um padrão de bits.

3. Qual o caracter que, na linguagem assembly do SPIM, inicia um comentário?

Resp: Letra A - #

4. Quantos bits há em cada instrução de máquina MIPS?

Resp: Letra C - 32

5. O que é o contador de programa?

Resp: Letra D - parte do processador que contém o endereço da próxima instrução de máquina para ser obtida.

6. Ao executarmos uma instrução, quanto será adicionado ao contador de programa?

Resp: Letra C - 4 (cada instrução ocupa 4 bytes)

7. O que é uma diretiva, tal como a diretiva .text?

Resp: Letra D - uma declaração que diz o montador algo sobre o que o programador quer, mas não corresponde diretamente a uma instrução de máquina. (diretivas são comandos para para customizar a compilação)

8. O que é um endereço simbólico?

Resp: Letra D - um nome usado no código-fonte em linguagem assembly para um local na memória.

9. Em qual endereço o simulador SPIM coloca a primeira instrução de máquina quando ele está sendo executado?

Resp: Letra B - 0x00400000

10. Algumas instruções de máquina possuem uma constante como um dos operandos. Como é chamado tal operando?

Resp: Letra A - operando imediato

11. Como é chamada uma operação lógica executada entre bits de cada coluna dos operandos para produzir um bit de resultado para cada coluna?

Resp: Letra B - operação bitwise (operação bit a bit)

12. Quando uma operação é de fato executada, como estão os operandos na ALU?

Resp: Letra D - cada um dos registradores deve possuir 32 bits.

13. Dezesesseis bits de dados de uma instrução de ori são usados como um operando imediato. Durante execução, o que deve ser feito primeiro?

Resp: Letra B - os dados são estendidos em zero à esquerda por 16 bits.

14. Qual das instruções seguintes armazenam no registrador \$5 um padrão de bits que representa positivo 48?

Resp: Letra C - ori \$5,\$0, 48

15. A instrução de ori pode armazenar o complemento de dois de um número em um registrador?

Resp: Letra A - Não (pois o ori não pode acrescentar 1 ao número, para virar complemento de dois).

16. Qual das instruções seguintes limpa todos os bits no registrador \$8 com exceção do byte de baixa ordem que fica inalterado?

Resp: Letra D - andi \$8,\$8,0xFF

17. Qual é o resultado de um ou exclusivo de padrão sobre ele mesmo?

Resp: Letra C - O padrão original utilizado

18. Todas as instruções de máquina têm os mesmos campos?

Resp: Letra A - Não. Diferentes de instruções de máquina possuem campos diferentes. (campos como Register, Imediato e Jump são diferentes um do outro)

Parte 2 – Programas MIPS/MARS

Programa 1 –

```
codigo_compilador_anotacoes.asm  p1.asm

1  #programa 1
2  # a = 2; b = 3; c = 4; d = 5;
3  # x = (a+b) - (c+d);
4  # y = a - b + x;
5  # b = x - y;
6
7  addi $s0, $zero, 2 # a
8  addi $s1, $zero, 3 # b
9  addi $s2, $zero, 4 # c
0  addi $s3, $zero, 5 # d
1
2  add $s4, $s0, $s1
3  add $s5, $s2, $s3
4  sub $s4, $s4, $s5 # x
5
6  sub $s5, $s2, $s3
7  add $s5, $s5, $s4 # y
8
9  sub $s1, $s4, $s5 # b
0
```

0x004000000x20100002addi \$16, \$0, 0x000000027: addi \$a0, \$zero, 2 # a

0x0040000040x20110003addi \$17, \$0, 0x000000038: addi \$a1, \$zero, 3 # b

0x0040000080x20120004addi \$18, \$0, 0x000000049: addi \$a2, \$zero, 4 # c

0x00400000c0x20130005addi \$19, \$0, 0x0000000510: addi \$a3, \$zero, 5 # d

0x0040000100x0211a020add \$20, \$16, \$1712: add \$a4, \$a5, \$a1

0x0040000140x0215a020add \$21, \$18, \$1913: add \$a5, \$a2, \$a3

0x0040000180x0219a020add \$22, \$19, \$2014: sub \$a4, \$a4, \$a5 # x

0x00400001c0x0253a022sub \$21, \$18, \$1916: sub \$a5, \$a2, \$a3

0x0040000200x02b4a020add \$21, \$20, \$2017: add \$a5, \$a5, \$a4 # y

0x0040000240x02355a022sub \$17, \$20, \$2119: sub \$a1, \$a5, \$a5 # b

0x100100000x0000000000x0000000000x0000000000x0000000000x0000000000x0000000000x0000000000x0000000000

0x100100020x0000000000x0000000000x0000000000x0000000000x0000000000x0000000000x0000000000x0000000000

0x100100040x0000000000x0000000000x0000000000x0000000000x0000000000x0000000000x0000000000x0000000000

0x100100060x0000000000x0000000000x0000000000x0000000000x0000000000x0000000000x0000000000x0000000000

0x100100080x0000000000x0000000000x0000000000x0000000000x0000000000x0000000000x0000000000x0000000000

0x1001000a0x0000000000x0000000000x0000000000x0000000000x0000000000x0000000000x0000000000x0000000000

0x1001000c0x0000000000x0000000000x0000000000x0000000000x0000000000x0000000000x0000000000x0000000000

0x1001000e0x0000000000x0000000000x0000000000x0000000000x0000000000x0000000000x0000000000x0000000000

0x100100100x0000000000x0000000000x0000000000x0000000000x0000000000x0000000000x0000000000x0000000000

0x100100120x0000000000x0000000000x0000000000x0000000000x0000000000x0000000000x0000000000x0000000000

0x100100140x0000000000x0000000000x0000000000x0000000000x0000000000x0000000000x0000000000x0000000000

0x100100160x0000000000x0000000000x0000000000x0000000000x0000000000x0000000000x0000000000x0000000000

0x100100180x0000000000x0000000000x0000000000x0000000000x0000000000x0000000000x0000000000x0000000000

0x1001001a0x0000000000x0000000000x0000000000x0000000000x0000000000x0000000000x0000000000x0000000000

0x1001000000x0000000000x0000000000x0000000000x0000000000x0000000000x0000000000x0000000000x0000000000

0x1001000200x0000000000x0000000000x0000000000x0000000000x0000000000x0000000000x0000000000x0000000000

0x1001000400x0000000000x0000000000x0000000000x0000000000x0000000000x0000000000x0000000000x0000000000

0x1001000600x0000000000x0000000000x0000000000x0000000000x0000000000x0000000000x0000000000x0000000000

0x1001000800x0000000000x0000000000x0000000000x0000000000x0000000000x0000000000x0000000000x0000000000

0x1001000a00x0000000000x0000000000x0000000000x0000000000x0000000000x0000000000x0000000000x0000000000

0x1001000c00x0000000000x0000000000x0000000000x0000000000x0000000000x0000000000x0000000000x0000000000

0x1001000e00x0000000000x0000000000x0000000000x0000000000x0000000000x0000000000x0000000000x0000000000

0x1001001000x0000000000x0000000000x0000000000x0000000000x0000000000x0000000000x0000000000x0000000000

0x1001001200x0000000000x0000000000x0000000000x0000000000x0000000000x0000000000x0000000000x0000000000

0x1001001400x0000000000x0000000000x0000000000x0000000000x0000000000x0000000000x0000000000x0000000000

0x1001001600x0000000000x0000000000x0000000000x0000000000x0000000000x0000000000x0000000000x0000000000

0x1001001800x0000000000x0000000000x0000000000x0000000000x0000000000x0000000000x0000000000x0000000000

0x1001001a00x0000000000x0000000000x0000000000x0000000000x0000000000x0000000000x0000000000x0000000000

0x1001000000x0000000000x0000000000x0000000000x0000000000x0000000000x0000000000x0000000000x0000000000

0x1001000200x0000000000x0000000000x0000000000x0000000000x0000000000x0000000000x0000000000x0000000000

0x1001000400x0000000000x0000000000x0000000000x0000000000x0000000000x0000000000x0000000000x0000000000

0x1001000600x0000000000x0000000000x0000000000x0000000000x0000000000x0000000000x0000000000x0000000000

0x1001000800x0000000000x0000000000x0000000000x0000000000x0000000000x0000000000x0000000000x0000000000

0x1001000a00x0000000000x0000000000x0000000000x0000000000x0000000000x0000000000x0000000000x0000000000

0x1001000c00x0000000000x0000000000x0000000000x0000000000x0000000000x0000000000x0000000000x0000000000

0x1001000e00x0000000000x0000000000x0000000000x0000000000x0000000000x0000000000x0000000000x0000000000

0x1001001000x0000000000x0000000000x0000000000x0000000000x0000000000x0000000000x0000000000x0000000000

0x1001001200x0000000000x0000000000x0000000000x0000000000x0000000000x0000000000x0000000000x0000000000

0x1001001400x0000000000x0000000000x0000000000x0000000000x0000000000x0000000000x0000000000x0000000000

0x1001001600x0000000000x0000000000x0000000000x0000000000x0000000000x0000000000x0000000000x0000000000

0x1001001800x0000000000x0000000000x0000000000x0000000000x0000000000x0000000000x0000000000x0000000000

0x1001001a00x0000000000x0000000000x0000000000x0000000000x0000000000x0000000000x0000000000x0000000000

0x1001000000x0000000000x0000000000x0000000000x0000000000x0000000000x0000000000x0000000000x0000000000

0x1001000200x0000000000x0000000000x0000000000x0000000000x0000000000x0000000000x0000000000x0000000000

0x1001000400x0000000000x0000000000x0000000000x0000000000x0000000000x0000000000x0000000000x0000000000

0x1001000600x0000000000x0000000000x0000000000x0000000000x0000000000x0000000000x0000000000x0000000000

0x1001000800x0000000000x0000000000x0000000000x0000000000x0000000000x0000000000x0000000000x0000000000

0x1001000a00x0000000000x0000000000x0000000000x0000000000x0000000000x0000000000x0000000000x0000000000

0x1001000c00x0000000000x0000000000x0000000000x0000000000x0000000000x0000000000x0000000000x0000000000

0x1001000e00x0000000000x0000000000x0000000000x0000000000x0000000000x0000000000x0000000000x0000000000

0x1001001000x0000000000x0000000000x0000000000x0000000000x0000000000x0000000000x0000000000x0000000000

0x1001001200x0000000000x0000000000x0000000000x0000000000x0000000000x0000000000x0000000000x0000000000

0x1001001400x0000000000x0000000000x0000000000x0000000000x0000000000x0000000000x0000000000x0000000000

0x1001001600x0000000000x0000000000x0000000000x0000000000x0000000000x0000000000x0000000000x0000000000

0x1001001800x0000000000x0000000000x0000000000x0000000000x0000000000x0000000000x0000000000x0000000000

0x1001001a00x0000000000x0000000000x0000000000x0000000000x0000000000x0000000000x0000000000x0000000000

RegistersCoproccCoprocc

NameNumberValue

\$zero00x00000000

\$at10x00000000

\$v020x00000000

\$v130x00000000

\$a040x00000000

\$a150x00000000

\$a260x00000000

\$a370x00000000

\$t080x00000000

\$t190x00000000

\$t2100x00000000

\$t3110x00000000

\$t4120x00000000

\$t5130x00000000

\$t6140x00000000

\$t7150x00000000

\$a0160x00000002

\$a1170x00000001

\$a2180x00000004

\$a3190x00000005

\$a4200xffffffff

\$a5210xffffffff

\$a6220x00000000

\$a7230x00000000

\$t0240x00000000

\$t1250x00000000

\$t2260x00000000

\$t3270x00000000

\$t4280x00000000

\$t5290x00000000

\$t6300x00000000

\$t7310x00000000

\$a0320x00000000

\$a1330x00000000

\$a2340x00000000

\$a3350x00000000

\$a4360x00000000

\$a5370x00000000

\$a6380x00000000

\$a7390x00000000

\$t0400x00000000

\$t1410x00000000

\$t2420x00000000

\$t3430x00000000

\$t4440x00000000

\$t5450x00000000

\$t6460x00000000

\$t7470x00000000

\$a0480x00000000

\$a1490x00000000

\$a2500x00000000

\$a3510x00000000

\$a4520x00000000

\$a5530x00000000

\$a6540x00000000

\$a7550x00000000

\$t0560x00000000

\$t1570x00000000

\$t2580x00000000

\$t3590x00000000

\$t4600x00000000

\$t5610x00000000

\$t6620x00000000

\$t7630x00000000

\$a0640x00000000

\$a1650x00000000

\$a2660x00000000

\$a3670x00000000

\$a4680x00000000

\$a5690x00000000

\$a6700x00000000

\$a7710x00000000

\$t0720x00000000

\$t1730x00000000

\$t2740x00000000

\$t3750x00000000

\$t4760x00000000

\$t5770x00000000

\$t6780x00000000

\$t7790x00000000

\$a0800x00000000

\$a1810x00000000

\$a2820x00000000

\$a3830x00000000

\$a4840x00000000

\$a5850x00000000

\$a6860x00000000

\$a7870x00000000

\$t0880x00000000

\$t1890x00000000

\$t2900x00000000

\$t3910x00000000

\$t4920x00000000

\$t5930x00000000

\$t6940x00000000

\$t7950x00000000

\$a0960x00000000

\$a1970x00000000

\$a2980x00000000

\$a3990x00000000

\$a41000x00000000

\$a51010x00000000

\$a61020x00000000

\$a71030x00000000

\$t01040x00000000

\$t11050x00000000

\$t21060x00000000

\$t31070x00000000

\$t41080x00000000

\$t51090x00000000

\$t61100x00000000

\$t71110x00000000

\$a01120x00000000

\$a11130x00000000

\$a21140x00000000

\$a31150x00000000

\$a41160x00000000

\$a51170x00000000

\$a61180x00000000

\$a71190x00000000

\$t01200x00000000

\$t11210x00000000

\$t21220x00000000

\$t31230x00000000

\$t41240x00000000

\$t51250x00000000

\$t61260x00000000

\$t71270x00000000

\$a01280x00000000

\$a11290x00000000

\$a21300x00000000

\$a31310x00000000

\$a41320x00000000

\$a51330x00000000

\$a61340x00000000

\$a71350x00000000

\$t01360x00000000

\$t11370x00000000

\$t21380x00000000

\$t31390x00000000

\$t41400x00000000

\$t51410x00000000

\$t61420x00000000

\$t71430x00000000

\$a01440x00000000

\$a11450x00000000

\$a21460x00000000

\$a31470x00000000

\$a41480x00000000

\$a51490x00000000

\$a61500x00000000

\$a71510x00000000

\$t01520x00000000

\$t11530x00000000

\$t21540x00000000

\$t31550x00000000

\$t41560x00000000

\$t51570x00000000

\$t61580x00000000

\$t71590x00000000

\$a01600x00000000

\$a11610x00000000

\$a21620x00000000

\$a31630x00000000

\$a41640x00000000

\$a51650x00000000

\$a61660x00000000

\$a71670x00000000

\$t01680x00000000

\$t11690x00000000

\$t21700x00000000

\$t31710x00000000

\$t41720x00000000

\$t51730x00000000

\$t61740x00000000

\$t71750x00000000

\$a01760x00000000

\$a11770x00000000

\$a21780x00000000

\$a31790x00000000

\$a41800x00000000

\$a51810x00000000

\$a61820x00000000

\$a71830x00000000

\$t01840x00000000

\$t11850x00000000

\$t21860x00000000

\$t31870x00000000

\$t41880x00000000

\$t51890x00000000

\$t61900x00000000

\$t71910x00000000

\$a01920x00000000

\$a11930x00000000

\$a21940x00000000

\$a31950x00000000

\$a41960x00000000

\$a51970x00000000

\$a61980x00000000

\$a71990x00000000

\$t02000x00000000

\$t12010x00000000

\$t22020x00000000

\$t32030x00000000

\$t42040x00000000

\$t52050x00000000

\$t62060x00000000

\$t72070x00000000

\$a02080x00000000

\$a12090x00000000

\$a22100x00000000

\$a32110x00000000

\$a42120x00000000

\$a52130x00000000

\$a62140x00000000

\$a72150x00000000

\$t02160x00000000

\$t12170x00000000

\$t22180x00000000

\$t32190x00000000

\$t42200x00000000

\$t52210x00000000

\$t62220x00000000

\$t72230x00000000

\$a02240x00000000

\$a12250x00000000

\$a22260x00000000

\$a32270x00000000

\$a42280x00000000

\$a52290x00000000

\$a62300x00000000

\$a72310x00000000

\$t02320x00000000

\$t12330x00000000

\$t22340x00000000

\$t32350x00000000

\$t42360x00000000

\$t52370x00000000

\$t62380x00000000

\$t72390x00000000

\$a02400x00000000

\$a12410x00000000

\$a22420x00000000

\$a32430x00000000

\$a42440x00000000

\$a52450x00000000

\$a62460x00000000

\$a72470x00000000

\$t02480x00000000

\$t12490x00000000

\$t22500x00000000

\$t32510x00000000

\$t42520x00000000

\$t52530x00000000

\$t62540x00000000

\$t72550x00000000

\$a02560x00000000

\$a12570x00000000

\$a22580x00000000

\$a32590x00000000

\$a42600x00000000

\$a52610x00000000

\$a62620x00000000

\$a72630x00000000

\$t02640x00000000

\$t12650x00000000

\$t22660x00000000

\$t32670x00000000

\$t42680x00000000

\$t52690x00000000

\$t62700x00000000

\$t72710x00000000

\$a02720x00000000

\$a12730x00000000

\$a22740x00000000

\$a32750x00000000

\$a42760x00000000

\$a52770x00000000

\$a62780x00000000

\$a72790x00000000

\$t02800x00000000

\$t12810x00000000

\$t22820x00000000

\$t32830x00000000

\$t42840x00000000

\$t52850x00000000

\$t62860x00000000

\$t72870x00000000

\$a02880x00000000

\$a12890x00000000

\$a22900x00000000

\$a32910x00000000

\$a42920x00000000

\$a52930x00000000

\$a62940x000000

Programa 2 –

Edit

Execute

codigo_compilador_annotacoes.asm

p1.asm

p2.asm

```

1  #programa 2
2  # x = 1;
3  # y = 5*x + 15;
4
5  addi $s0, $zero, 1
6  add $s1, $s0, $s0 # y = 1 + 1 = 2
7  add $s1, $s1, $s1 # y = 2 + 2 = 4
8  add $s1, $s1, $s0 # y = 4 + 1 = 5
9
10 addi $s1, $s1, 15 # y = 5 + 15 = 20

```

Text Segment

Byte	Address	Code	Basic	Source
	0x00400000	0x23108001	addi \$t0,\$zero,1	5: addi \$t0, \$zero, 1
	0x00400004	0x02108820	add \$t1,\$t0,\$t0	6: add \$t1, \$t0, \$t0 # y = 1 + 1 = 2
	0x00400008	0x02318820	add \$t1,\$t1,\$t1	7: add \$t1, \$t1, \$t1 # y = 2 + 2 = 4
	0x0040000c	0x02308820	add \$t1,\$t1,\$t0	8: add \$t1, \$t1, \$t0 # y = 4 + 1 = 5
	0x00400010	0x2331000f	addi \$t1,\$t1,15	10: addi \$t1, \$t1, 15 # y = 5 + 15 = 20

Data Segment

Address	Value (+0)	Value (+4)	Value (+8)	Value (+c)	Value (+10)	Value (+14)	Value (+18)	Value (+1c)
0x10010000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x10010020	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x10010040	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x10010060	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x10010080	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x100100a0	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x100100c0	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x100100e0	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x10010100	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x10010120	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x10010140	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x10010160	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x10010180	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x100101a0	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000

Registers

Coproc 1

Coproc 0

Name	Number	Value
\$zero	0	0x00000000
\$at	1	0x00000000
\$v0	2	0x00000000
\$v1	3	0x00000000
\$a0	4	0x00000000
\$a1	5	0x00000000
\$a2	6	0x00000000
\$a3	7	0x00000000
\$t0	8	0x00000000
\$t1	9	0x00000000
\$t2	10	0x00000000
\$t3	11	0x00000000
\$t4	12	0x00000000
\$t5	13	0x00000000
\$t6	14	0x00000000
\$t7	15	0x00000000
\$s0	16	0x00000001
\$s1	17	0x00000014
\$s2	18	0x00000000
\$s3	19	0x00000000
\$s4	20	0x00000000
\$s5	21	0x00000000
\$s6	22	0x00000000
\$s7	23	0x00000000
\$s8	24	0x00000000
\$s9	25	0x00000000
\$k0	26	0x00000000
\$k1	27	0x00000000
\$fp	28	0x10000000
\$gp	29	0x7ffffcfc
\$fp	30	0x00000000
\$ra	31	0x00000000
\$pc		0x00000014
\$lo		0x00000000

Programa 3 –

Edit

Execute

codigo_compilador_annotacoes.asm

p1.asm

p2.asm

p3.asm

```

1  #programa 3
2  # x = 3; y = 4;
3  # z = ( 15*x + 67*y)*4
4
5  addi $s0, $zero, 3 # x = 3
6  addi $s1, $zero, 4 # y = 4
7
8  add $t0, $s0, $s0 # t0 = 3 + 3 = 6
9  add $t0, $t0, $t0 # t0 = 6 + 6 = 12
10 add $t0, $t0, $t0 # t0 = 12 + 12 = 24
11 add $t0, $t0, $t0 # t0 = 24 + 24 = 48
12 sub $t0, $t0, $s0 # t0 = 48 - 3 = 45 (15 * 3)
13
14 add $t1, $s1, $s1 # t1 = 4 + 4 = 8
15 add $t1, $t1, $t1 # t1 = 8 + 8 = 16
16 add $t1, $t1, $t1 # t1 = 16 + 16 = 32
17 add $t1, $t1, $t1 # t1 = 32 + 32 = 64
18 add $t1, $t1, $t1 # t1 = 64 + 64 = 128
19 add $t1, $t1, $t1 # t1 = 128 + 128 = 256
20 add $t1, $t1, $s1 # t1 = 256 + 4 = 260
21 add $t1, $t1, $s1 # t1 = 260 + 4 = 264
22 add $t1, $t1, $s1 # t1 = 264 + 4 = 268
23
24 add $s2, $t0, $t1 # z = 268 + 45
25 add $s2, $s2, $s2 # z = z + z (2z)
26 add $s2, $s2, $s2 # z = 2z + 2z
27

```

Edit Execute					Registers Coproc 1 Coproc 0			
Text Segment					Name	Number	Value	
Inst	Address	Code	Basic	Source				
☐	0x00400000	0x20100003	addi \$t6,\$0,0x00000003	5: addi \$a0, \$zero, 3 # x = 3	\$zero	0	0x00000000	
☐	0x00400004	0x20100004	addi \$t7,\$0,0x00000004	6: addi \$a1, \$zero, 4 # y = 4	\$at	1	0x00000000	
☐	0x00400008	0x02040020	add \$t8,\$t6,\$t6	8: add \$t0, \$a0, \$a0 # t0 = 3 + 3 = 6	\$v0	2	0x00000000	
☐	0x0040000c	0x01040020	add \$t9,\$t8,\$t8	9: add \$t0, \$t0, \$t0 # t0 = 6 + 6 = 12	\$v1	3	0x00000000	
☐	0x00400010	0x01040020	add \$t9,\$t8,\$t8	10: add \$t0, \$t0, \$t0 # t0 = 12 + 12 = 24	\$a0	4	0x00000000	
☐	0x00400014	0x01040020	add \$t9,\$t8,\$t8	11: add \$t0, \$t0, \$t0 # t0 = 24 + 24 = 48	\$a1	5	0x00000000	
☐	0x00400018	0x01140022	sub \$t8,\$t8,\$t6	12: sub \$t0, \$t0, \$a0 # t0 = 48 - 3 = 45 (15 * 3)	\$a2	6	0x00000000	
☐	0x0040001c	0x02140020	add \$t9,\$t17,\$t17	14: add \$t1, \$t1, \$t1 # t1 = 4 + 4 = 8	\$a3	7	0x00000000	
☐	0x00400020	0x02140020	add \$t9,\$t9,\$t9	15: add \$t1, \$t1, \$t1 # t1 = 8 + 8 = 16	\$t0	8	0x0000002d	
☐	0x00400024	0x02140020	add \$t9,\$t9,\$t9	16: add \$t1, \$t1, \$t1 # t1 = 16 + 16 = 32	\$t1	9	0x0000001c	
☐	0x00400028	0x02140020	add \$t9,\$t9,\$t9	17: add \$t1, \$t1, \$t1 # t1 = 32 + 32 = 64	\$t2	10	0x00000000	
☐	0x0040002c	0x02140020	add \$t9,\$t9,\$t9	18: add \$t1, \$t1, \$t1 # t1 = 64 + 64 = 128	\$t3	11	0x00000000	
☐	0x00400030	0x02140020	add \$t9,\$t9,\$t9	19: add \$t1, \$t1, \$t1 # t1 = 128 + 128 = 256	\$t4	12	0x00000000	
☐	0x00400034	0x03140020	add \$t9,\$t9,\$t17	20: add \$t1, \$t1, \$a1 # t1 = 256 + 4 = 260	\$t5	13	0x00000000	
☐	0x00400038	0x03140020	add \$t9,\$t9,\$t17	21: add \$t1, \$t1, \$a1 # t1 = 260 + 4 = 264	\$t6	14	0x00000000	
☐	0x0040003c	0x03140020	add \$t9,\$t9,\$t17	22: add \$t1, \$t1, \$a1 # t1 = 264 + 4 = 268	\$t7	15	0x00000000	
☐	0x00400040	0x01040020	add \$t18,\$t8,\$t8	24: add \$a2, \$a0, \$a1 # a2 = 268 + 45	\$a0	16	0x00000003	
					\$a1	17	0x00000004	
					\$a2	18	0x00000045	
					\$a3	19	0x00000000	
					\$a4	20	0x00000000	
					\$a5	21	0x00000000	
					\$a6	22	0x00000000	
					\$a7	23	0x00000000	
					\$a8	24	0x00000000	
					\$t9	25	0x00000000	
					\$t0	26	0x00000000	
					\$t1	27	0x00000000	
					\$t2	28	0x00000000	
					\$t3	29	0xffffffff	
					\$t4	30	0x00000000	
					\$t5	31	0x00000000	
					\$t6		0x00000000	
					\$t7		0x00000000	
					\$t8		0x00000000	
					\$t9		0x00000000	
					\$t0		0x00000000	
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					\$t6		0x00000000	
					\$t7		0x00000000	
					\$t8		0x00000000	
					\$t9		0x00000000	
					\$t0		0x00000000	
					\$t1		0x00000000	
					\$t2		0x00000000	
					\$t3		0x00000000	
					\$t4			

Programa 5 –

Edit Execute

p4.asm

p5.asm

```

1  # x = 100000;
2  # y = 200000;
3  # z = x + y;
4
5  ori $t0, $zero, 0x186A
6  sll $s0, $t0, 4 # 6250 * 16(2^4) = 100000
7
8
9  ori $t1, $zero, 0x30D4
10 sll $s1, $t1, 4 # 12500 16(2^4) = 200000
11
12 add $s2, $s0, $s1 # x + y

```

Edit Execute

Offset	Address	Code	Basic	Source
0x00400000	0x3408186A	ori \$t0, \$zero, 0x186A	5: ori \$t0, \$zero, 0x186A	
0x00400004	0x00081004	sll \$s0, \$t0, 4 # 6250 * 16(2^4) = 100000	6: sll \$s0, \$t0, 4 # 6250 * 16(2^4) = 100000	
0x00400008	0x000830D4	ori \$t1, \$zero, 0x30D4	9: ori \$t1, \$zero, 0x30D4	
0x0040000C	0x00085900	sll \$s1, \$t1, 4 # 12500 16(2^4) = 200000	10: sll \$s1, \$t1, 4 # 12500 16(2^4) = 200000	
0x00400010	0x02119020	add \$s2, \$s0, \$s1 # x + y	12: add \$s2, \$s0, \$s1 # x + y	

Edit Execute

Address	Value (+0)	Value (+4)	Value (+8)	Value (+C)	Value (+10)	Value (+14)	Value (+18)	Value (+1C)
0x10010000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x10010020	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x10010040	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x10010060	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x10010080	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x100100A0	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x100100C0	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x100100E0	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x10010100	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x10010120	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x10010140	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x10010160	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x10010180	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x100101A0	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000

Registers Coproc 1 Coproc 0

Name	Number	Value
\$zero	0	0x00000000
\$at	1	0x00000000
\$v0	2	0x00000000
\$v1	3	0x00000000
\$a0	4	0x00000000
\$a1	5	0x00000000
\$a2	6	0x00000000
\$a3	7	0x00000000
\$t0	8	0x0000186A
\$t1	9	0x000030D4
\$t2	10	0x00000000
\$t3	11	0x00000000
\$t4	12	0x00000000
\$t5	13	0x00000000
\$t6	14	0x00000000
\$t7	15	0x00000000
\$s0	16	0x00001860
\$s1	17	0x00003040
\$s2	18	0x000493e0
\$s3	19	0x00000000
\$s4	20	0x00000000

Programa 6 –

Edit Execute

p4.asm p5.asm p6.asm*

```

1  #programa 6
2
3  ori $t0, $zero, 0x7FFF
4  sll $t0, $t0, 16
5  ori $s0, $t0, 0xFFFF # maior numero possivel
6
7  ori $t1, $zero, 0x493E
8  sll $s1, $t1, 4 # 300.000
9
10 sll $t2, $s1, 2 # 1.200.000
11 sub $s2, $s0, $t2

```

Edit Execute

☐ Text Segment

Byte	Address	Code	Basic	Source
	0x00400000	0x34087fff	ori \$t0, \$zero, 0x7FFF	3: ori \$t0, \$zero, 0x7FFF
	0x00400004	0x00084400	sll \$t0, \$t0, 16	4: sll \$t0, \$t0, 16
	0x00400008	0x3510ffff	ori \$s0, \$t0, 0xFFFF	5: ori \$s0, \$t0, 0xFFFF # maior numero possivel
	0x0040000c	0x3409493e	ori \$t1, \$zero, 0x493E	7: ori \$t1, \$zero, 0x493E
	0x00400010	0x00089800	sll \$s1, \$t1, 4	8: sll \$s1, \$t1, 4 # 300.000
	0x00400014	0x00158000	sll \$t2, \$s1, 2	10: sll \$t2, \$s1, 2 # 1.200.000
	0x00400018	0x020a9020	sub \$s2, \$s0, \$t2	11: sub \$s2, \$s0, \$t2

☐ Data Segment

Address	Value (-0)	Value (+4)	Value (+8)	Value (+c)	Value (+10)	Value (+14)	Value (+18)	Value (+1c)
0x10010000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x10010020	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x10010040	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x10010060	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x10010080	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x100100a0	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x100100c0	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x100100e0	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x10010100	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x10010120	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x10010140	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x10010160	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x10010180	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x100101a0	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000

☐ Registers
 Coproc 1
 Coproc 0

☐ Coproc 1
 Coproc 0

Name	Number	Value
\$zero	0	0x00000000
\$at	1	0x00000000
\$v0	2	0x00000000
\$v1	3	0x00000000
\$a0	4	0x00000000
\$a1	5	0x00000000
\$a2	6	0x00000000
\$a3	7	0x00000000
\$t0	8	0x7fffffff
\$t1	9	0x0000493e
\$t2	10	0x00124800
\$t3	11	0x00000000
\$t4	12	0x00000000
\$t5	13	0x00000000
\$t6	14	0x00000000
\$t7	15	0x00000000
\$s0	16	0x7fffffff
\$s1	17	0x0000493e
\$s2	18	0x7fffe077
\$s3	19	0x00000

Programa 7 –

Edit
Execute

p4.asm
p5.asm
p6.asm
p7.asm

```

1  #programa 7
2  #devemos chegar a sem add ou sub
3  # $8 == $t0
4
5  ori $8, $zero, 0x01
6  sll $t0, $t0, 31
7  sra $t0, $t0, 31
8

```

Edit
Execute

Text Segment

Byte	Address	Code	Basic	Source
	0x00400000	0x34080001	ori \$8,\$0,0x00000001	5: ori \$8, \$zero, 0x01
	0x00400004	0x000847c0	sll \$t0,\$t0,31	6: sll \$t0, \$t0, 31
	0x00400008	0x000847c3	sra \$t0,\$t0,31	7: sra \$t0, \$t0, 31

Data Segment

Address	Value (+0)	Value (+4)	Value (+8)	Value (+c)	Value (+10)	Value (+14)	Value (+18)	Value (+1c)
0x10010000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x10010004	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x10010008	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x1001000c	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x10010010	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x10010014	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x10010018	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x1001001c	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x10010020	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000

Registers
Coproc 1
Coproc 0

Name	Number	Value
\$zero	0	0x00000000
\$at	1	0x00000000
\$v0	2	0x00000000
\$v1	3	0x00000000
\$a0	4	0x00000000
\$a1	5	0x00000000
\$a2	6	0x00000000
\$a3	7	0x00000000
\$t0	8	0xffffffff
\$t1	9	0x00000000
\$t2	10	0x00000000
\$t3	11	0x00000000
\$t4	12	0x00000000
\$t5	13	0x00000000
\$t6	14	0x00000000
\$t7	15	0x00000000
\$s0	16	0x00000000
\$s1	17	0x00000000
\$s2	18	0x00000000
\$s3	19	0x00000000
\$s4	20	0x00000000
\$s5	21	0x00000000
\$s6	22	0x00000000
\$s7	23	0x00000000
\$s8	24	0x00000000
\$s9	25	0x00000000
\$k0	26	0x00000000
\$k1	27	0x00000000
\$gp	28	0x10000000
\$sp	29	0x7ffffcfc
\$fp	30	0x00000000
\$ra	31	0x00000000
\$pc		0x00000000
\$hi		0x00000000
\$lo		0x00000000

Programa 8 –

Edit
Execute

p4.asm
p5.asm
p6.asm
p7.asm
p8.asm

```

1  #programa 7
2
3  ori $8, $zero, 0x1234
4  sll $8, $8, 16
5  ori $8, $8, 0x5678
6
7  srl $8, $9, 24 #0x12
8
9  sll $10, $8, 8
10 srl $10, $10, 24 #0x34
11
12 andi $11, $8, 0xFF00
13 srl $11, $11, 8 #0x56
14
15 andi $12, $8, 0x00FF #0x78
16

```

Text Segment

Byte	Address	Code	Basic	Source
	0x00400000	0x34081234	ori \$8,\$0,0x00001234	3: ori \$8, \$zero, 0x1234
	0x00400004	0x00084400	sll \$8,\$8,0x00000010	4: sll \$8, \$8, 16
	0x00400008	0x35085678	ori \$8,\$8,0x00005678	5: ori \$8, \$8, 0x5678
	0x0040000c	0x00094602	srl \$8,\$9,0x00000018	7: srl \$8, \$9, 24 #0x12
	0x00400010	0x00085200	sll \$10,\$8,8	9: sll \$10, \$8, 8
	0x00400014	0x00084502	srl \$10,\$10,0x00000018	10: srl \$10, \$10, 24 #0x34
	0x00400018	0x310bf000	andi \$11,\$8,0x0000ff00	12: andi \$11, \$8, 0xFF00
	0x0040001c	0x000b5a02	srl \$11,\$11,8	13: srl \$11, \$11, 8 #0x56
	0x00400020	0x310c00ff	andi \$12,\$8,0x000000ff	15: andi \$12, \$8, 0x00FF #0x78

Data Segment

Address	Value (+0)	Value (+4)	Value (+8)	Value (+c)	Value (+10)	Value (+14)	Value (+18)	Value (+1c)
0x10010000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x10010004	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x10010008	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x1001000c	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x10010010	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x10010014	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x10010018	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x1001001c	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x10010020	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000

Registers
Coproc 1
Coproc 0

Name	Number	Value
\$zero	0	0x00000000
\$at	1	0x00000000
\$v0	2	0x00000000
\$v1	3	0x00000000
\$a0	4	0x00000000
\$a1	5	0x00000000
\$a2	6	0x00000000
\$a3	7	0x00000000
\$t0	8	0x00000000
\$t1	9	0x00000000
\$t2	10	0x00000000
\$t3	11	0x00000000
\$t4	12	0x00000000
\$t5	13	0x00000000
\$t6	14	0x00000000
\$t7	15	0x00000000
\$s0	16	0x00000000
\$s1	17	0x00000000
\$s2	18	0x00000000
\$s3	19	0x00000000
\$s4	20	0x00000000
\$s5	21	0x00000000
\$s6	22	0x00000000
\$s7	23	0x00000000
\$s8	24	0x00000000
\$s9	25	0x00000000
\$k0	26	0x00000000
\$k1	27	0x00000000
\$gp	28	0x10000000
\$sp	29	0x7ffffcfc
\$fp	30	0x00000000
\$ra	31	0x00000000
\$pc		0x00400024
\$hi		0x00000000
\$lo		0x00000000

Programa 9 –

```

Edit Execute
p8.asm p9.asm* p5.asm
1 #programa 9
2
3 .data
4 x1: .word 15 # o 15 esta sendo salvo no endereco 0x10010000
5 x2: .word 25 # salvo no endereco 0x10010004
6 x3: .word 13 # salvo no endereco 0x10010008
7 x4: .word 17 # salvo no endereco 0x1001000c
8 soma: .word -1 # salvo no endereco 0x10010010
9
10 .text
11 ori $t0, $zero, 0x1001 #0x00001001
12 sll $t0, $t0, 16 # shift de 16 bits -> 0x10010000
13
14 lw $s0, 0($t0) # salva o que esta no endereco t0 + 0 (0x10010000), que guarda o 15, no $s0
15 lw $s1, 4($t0) # salva o que esta no endereco t0 + 4 (0x10010004), que guarda o 25, no $s1
16 lw $s2, 8($t0)
17 lw $s3, 0x000c($t0) #12
18
19 add $t1, $s0, $s1 #t1 = x1 + x2
20 add $t1, $t1, $s2
21 add $s4, $t1, $s3 # soma todos os registradores
22
23 sw $s4, 16($t0) # armazena $s4 (70) no endereco 0x10010010 (soma)
24

```

0x40000000

ori \$t0, \$zero, 0x00001001

11: ori \$t0, \$zero, 0x00001001

0x40000004

sll \$t0, \$t0, 16 # shift de 16 bits -> 0x10010000

12: sll \$t0, \$t0, 16 # shift de 16 bits -> 0x10010000

0x40000008

lw \$s0, 0(\$t0) # salva o que esta no endereco t0 + 0 (0x10010000), que guarda o 15, no \$s0

14: lw \$s0, 0(\$t0) # salva o que esta no endereco t0 + 0 (0x10010000), que guarda o 15, no \$s0

0x4000000c

lw \$s1, 4(\$t0) # salva o que esta no endereco t0 + 4 (0x10010004), que guarda o 25, no \$s1

15: lw \$s1, 4(\$t0) # salva o que esta no endereco t0 + 4 (0x10010004), que guarda o 25, no \$s1

0x40000010

lw \$s2, 8(\$t0)

16: lw \$s2, 8(\$t0)

0x40000014

lw \$s3, 0x000c(\$t0) #12

17: lw \$s3, 0x000c(\$t0) #12

0x40000018

add \$t1, \$s0, \$s1 #t1 = x1 + x2

19: add \$t1, \$s0, \$s1 #t1 = x1 + x2

0x4000001c

add \$t1, \$t1, \$s2

20: add \$t1, \$t1, \$s2

0x40000020

add \$s4, \$t1, \$s3 # soma todos os registradores

21: add \$s4, \$t1, \$s3 # soma todos os registradores

0x40000024

sw \$s4, 16(\$t0) # armazena \$s4 (70) no endereco 0x10010010 (soma)

23: sw \$s4, 16(\$t0) # armazena \$s4 (70) no endereco 0x10010010 (soma)

Registers

Coproc 0

Coproc 0

Name	Number	Value
\$zero	0	0x00000000
\$at	1	0x00000000
\$v0	2	0x00000000
\$v1	3	0x00000000
\$a0	4	0x00000000
\$a1	5	0x00000000
\$a2	6	0x00000000
\$a3	7	0x00000000
\$t0	8	0x10010000
\$t1	9	0x00000015
\$t2	10	0x00000000
\$t3	11	0x00000000
\$t4	12	0x00000000
\$t5	13	0x00000000
\$t6	14	0x00000000
\$t7	15	0x00000000
\$t8	16	0x00000000
\$t9	17	0x00000000
\$s0	18	0x00000000
\$s1	19	0x00000000
\$s2	20	0x00000000
\$s3	21	0x00000000
\$s4	22	0x00000000
\$s5	23	0x00000000
\$s6	24	0x00000000
\$s7	25	0x00000000
\$s8	26	0x00000000
\$s9	27	0x00000000
\$gp	28	0x10008000
\$sp	29	0x7ffffc00
\$fp	30	0x00000000
\$ra	31	0x00000000
pc	0x40000028	0x00000000
hi	0x00000000	0x00000000
lo	0x00000000	0x00000000

0x10010000

0x00000000

0x00000000

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EditExecute

Text Segment

Byte	Address	Code	Basic	Source
☐	0x00400000	0x34081001	ori \$t0,\$zero,0x1001	10: ori \$t0, \$zero, 0x1001
☐	0x00400004	0x00044400	all \$t0,\$t0,16	11: sll \$t0, \$t0, 16 #shift de 16 bits -> 0x10010000
☐	0x00400008	0x8d100000	lw \$s0,0(\$t0)	13: lw \$s0, 0(\$t0) # \$s0 = 5
☐	0x0040000c	0xd1100040	lw \$t1,4(\$t0)	14: lw \$t1, 4(\$t0) # \$t1 = 7
☐	0x00400010	0x001049c0	all \$t1,\$t1,7	16: sll \$t1, \$t1, 7 # 5 * 2^7 = 640
☐	0x00400014	0x01304222	sub \$t0,\$t1,640	17: sub \$t1, \$t1, \$t0 # 640 - 5 = 635 (125 * 5)
☐	0x00400018	0x00115180	all \$t0,\$t0,64	19: sll \$t0, \$t0, 6 # 7 * 64
☐	0x0040001c	0x15150000	add \$t0,\$t0,64	20: add \$t2, \$t0, \$t1 # (7 * 64) + 5
☐	0x00400020	0x012a9022	sub \$t0,\$t2,\$t0	22: sub \$s2, \$t1, \$t2 # y
☐	0x00400024	0x22520001	addi \$t0,\$t0,1	23: addi \$s2, \$s2, 1
☐	0x00400028	0xd1200008	sw \$s2,8(\$t0)	25: sw \$s2, 8(\$t0) #armazena o resultado no y da memoria

Data Segment

Address	Value (+0)	Value (+4)	Value (+8)	Value (+c)	Value (+10)	Value (+14)	Value (+18)	Value (+1c)
0x10010000	0x00000005	0x00000007	0x0000000b	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x10010020	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x10010040	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x10010060	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x10010080	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x100100a0	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x100100c0	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x100100e0	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x10010100	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x10010120	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x10010140	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x10010160	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x10010180	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x100101a0	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000

Registers

Name	Number	Value
\$zero	0	0x00000000
\$at	1	0x00000000
\$v0	2	0x00000000
\$v1	3	0x00000000
\$a0	4	0x00000000
\$a1	5	0x00000000
\$a2	6	0x00000000
\$a3	7	0x00000000
\$t0	8	0x10010000
\$t1	9	0x0000001b
\$t2	10	0x0000001c
\$t3	11	0x00000000
\$t4	12	0x00000000
\$t5	13	0x00000000
\$t6	14	0x00000000
\$t7	15	0x00000000
\$s0	16	0x00000005
\$s1	17	0x00000007
\$s2	18	0x0000000b
\$s3	19	0x00000000
\$s4	20	0x00000000
\$s5	21	0x00000000
\$s6	22	0x00000000
\$s7	23	0x00000000
\$s8	24	0x00000000
\$s9	25	0x00000000
\$k0	26	0x00000000
\$k1	27	0x00000000
\$gp	28	0x10008000
\$sp	29	0xffffffff
\$fp	30	0x00000000
\$ra	31	0x00000000
\$pc		0x00400028
\$hi		0x00000000
\$lo		0x00000000

Programa 11 –

EditExecute

p8.asm

p9.asm

p5.asm

p10.asm

p11.asm*

```
1 #programa 11
2 # y = x - z + 300000
3
4 .data
5 x: .word 100000
6 z: .word 200000
7 y: .word 0 # esse valor deverá ser sobrescrito após a execução do programa
8
9 .text
10 ori $t0, $zero, 0x1001
11 sll $t0, $t0, 16 #shift de 16 bits -> 0x10010000
12
13 ori $t1, $zero, 0x493E # Carrega o valor 18750 em $t1
14 sll $t1, $t1, 4 # 300000
15
16 lw $s0, 0($t0) # $s0 = 100000
17 lw $s1, 4($t0) # $s1 = 200000
18
19 sub $s2, $s0, $s1
20 add $s2, $s2, $t1 #resultado
21
22 sw $s2, 8($t0) #armazena o resultado no y da memoria
```

EditExecute

Text Segment

Byte	Address	Code	Basic	Source
☐	0x00400000	0x34081001	ori \$t0,\$zero,0x1001	10: ori \$t0, \$zero, 0x1001
☐	0x00400004	0x00044400	all \$t0,\$t0,16	11: sll \$t0, \$t0, 16 #shift de 16 bits -> 0x10010000
☐	0x00400008	0x3408493e	ori \$t1,\$zero,0x493e	13: ori \$t1, \$zero, 0x493E # Carrega o valor 18750 em \$t1
☐	0x0040000c	0x00049000	all \$t1,\$t1,4	14: sll \$t1, \$t1, 4 # 300000
☐	0x00400010	0x8d100000	lw \$s0,0(\$t0)	16: lw \$s0, 0(\$t0) # \$s0 = 100000
☐	0x00400014	0x8d110040	lw \$s1,4(\$t0)	17: lw \$s1, 4(\$t0) # \$s1 = 200000
☐	0x00400018	0x02118002	sub \$s2,\$s0,\$s1	19: sub \$s2, \$s0, \$s1
☐	0x0040001c	0x02499020	add \$s2,\$s2,\$t1	20: add \$s2, \$s2, \$t1 #resultado
☐	0x00400020	0xd1200008	sw \$s2,8(\$t0)	22: sw \$s2, 8(\$t0) #armazena o resultado no y da memoria

Data Segment

Address	Value (+0)	Value (+4)	Value (+8)	Value (+c)	Value (+10)	Value (+14)	Value (+18)	Value (+1c)
0x10010000	0x000156a0	0x000303d0	0x000303d0	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x10010020	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x10010040	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x10010060	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x10010080	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x100100a0	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x100100c0	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x100100e0	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x10010100	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x10010120	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x10010140	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x10010160	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x10010180	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x100101a0	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000

Registers

Name	Number	Value
\$zero	0	0x00000000
\$at	1	0x00000000
\$v0	2	0x00000000
\$v1	3	0x00000000
\$a0	4	0x00000000
\$a1	5	0x00000000
\$a2	6	0x00000000
\$a3	7	0x00000000
\$t0	8	0x10010000
\$t1	9	0x000493e0
\$t2	10	0x00000000
\$t3	11	0x00000000
\$t4	12	0x00000000
\$t5	13	0x00000000
\$t6	14	0x00000000
\$t7	15	0x00000000
\$s0	16	0x000156a0
\$s1	17	0x000303d0
\$s2	18	0x000303d0
\$s3	19	0x00000000
\$s4	20	0x00000000
\$s5	21	0x00000000
\$s6	22	0x00000000
\$s7	23	0x00000000
\$s8	24	0x00000000
\$s9	25	0x00000000
\$k0	26	0x00000000
\$k1	27	0x00000000
\$gp	28	0x10008000
\$sp	29	0xffffffff
\$fp	30	0x00000000
\$ra	31	0x00000000
\$pc		0x00400020
\$hi		0x00000000
\$lo		0x00000000

Programa 12 –

p8.asm	p9.asm	p5.asm	p10.asm	p11.asm	p12.asm
1	#programa 12				
2					
3	.data				
4	k: .word endl # 0x10010000 -> 0x10010004				
5	endl: .word x # 0x10010004 -> 0x10010008				
6	x: .word 5 # 0x10010008 -> 5				
7					
8	.text				
9	ori \$t0, \$zero, 0x1001				
10	sll \$t0, \$t0, 16 #shift de 16 bits -> 0x10010000				
11					
12	lw \$s0, 8(\$t0) # \$s0 = 5				
13					
14	sll \$s0, \$s0, 1				
15					
16	sw \$s0, 0(\$t0)				

Registers Coproc 1 Coproc 0

☐ Text Segment

Bbpt	Address	Code	Basic	Source
☐	0x00400000	0x34081001	ori \$8,\$0,0x00001001	9: ori \$t0,\$zero,0x1001
☐	0x00400004	0x00944000	sl \$8,\$8,0x00000010	10: sl \$t0,\$t0,16 #shift de 16 bits -> 0x10010000
☐	0x00400008	0x34120008	lw \$16,0x00000008(\$8)	11: lw \$a0,\$t0 # \$a0 = \$
☐	0x0040000c	0x00108040	sl \$16,\$16,0x00000001	14: sl \$a0,\$t0,1
☐	0x00400010	0xad100000	sw \$16,0x00000000(\$8)	16: sw \$a0,0(\$t0)

Name	Number	Value
\$zero	0	0x00000000
\$at	1	0x00000000
\$v0	2	0x00000000
\$v1	3	0x00000000
\$a0	4	0x00000000
\$a1	5	0x00000000
\$a2	6	0x00000000
\$a3	7	0x00000000
\$t0	8	0x10010000
\$t1	9	0x00000000
\$t2	10	0x00000000
\$t3	11	0x00000000
\$t4	12	0x00000000
\$t5	13	0x00000000
\$t6	14	0x00000000
\$t7	15	0x00000000
\$a0	16	0x00000000
\$a1	17	0x00000000
\$a2	18	0x00000000
\$a3	19	0x00000000
\$a4	20	0x00000000
\$a5	21	0x00000000
\$a6	22	0x00000000
\$a7	23	0x00000000
\$t8	24	0x00000000
\$t9	25	0x00000000
\$d0	26	0x00000000
\$d1	27	0x00000000
\$gp	28	0x10000000
\$sp	29	0xffffffff
\$fp	30	0x00000000
\$ra	31	0x00000000
pc		0x00400001
hi		0x00000000
lo		0x00000000

Address	Value (-D)	Value (-I)	Value (-B)	Value (-C)	Value (-10)	Value (-14)	Value (-18)	Value (-1c)
0x10010000	0x0000000a	0x10010008	0x00000005	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x10010020	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x10010040	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x10010060	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x10010080	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x100100A0	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x100100C0	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x100100E0	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x10010100	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x10010120	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x10010140	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x10010160	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x10010180	0x00000000	0x00000000	0					

Programa 13 –

```

1 #programa 13
2
3 .data
4 a: .word -100
5
6 .text
7 ori $t0, $zero, 1
8
9 # lui $t1, 0x1001 faz isso:
10 ori $t1, $zero, 0x1001
11 sll $t1, $t1, 16
12
13 lw $t3, 0($t1)
14
15 srl $t2, $t3, 31 # isola o bit de sinal
16
17 beq $t0, $t2, negativo # se o bit de sinal e 1: vai p/ negativo p/ encontrar seu modulo
18 j escrita # se nao for, pula pra escrita
19
20 negativo:
21 sub $t3, $zero, $t3 # -(-x) = x
22
23 escrita:
24 sw $t3 0($t1)

```

Bpt	Address	Code	Basic	Source
[x]	0x0400000	0x34080001	ori \$t0,\$zero, 1	7: ori \$t0, \$zero, 1
[x]	0x0400004	0xf09f0101	ori \$t1,\$zero, 0xc001	10: ori \$t1, \$zero, 0xc001
[x]	0x0400008	0x0094c0e1	\$l \$r, \$zero, 0x00000010	11: sll \$t1, \$t1, 16
[x]	0x040000c	0xd42b0000	lw \$t1, 0(x0)	13: lw \$t3, 0(\$t1)
[x]	0x0400010	0x000b57c2	srl \$t2,\$t3, 31 # isolia o bit de sinal	15: srl \$t2, \$t3, 31 # isolia o bit de sinal
[x]	0x0400014	0x11ae001b	bcs \$t0,\$t1, 0x00000001	17: bcs (\$t), \$t3, negativo # se o bit de sinal e 1: val p/ negativo p/ encontrar seu modulo
[x]	0x0400018	0x08100008j	0x00400020	18; j escrita t as nao for, pula pra escrita
[x]	0x040001c	0x000b5822	sub \$t1,\$t0,\$t1	21: sub \$t3, \$zero, \$t3 -(-\$x) = x
[x]	0x0400020	0xad2b0000	sw \$t1, 0(x0)	24: sw \$t3 0(\$t1)

Address	Value (+0)	Value (+4)	Value (+8)	Value (+C)	Value (+10)	Value (+14)	Value (+18)	Value (+1C)
0x10010000	0x0000000E4	0x000000000	0x000000000	0x000000000	0x000000000	0x000000000	0x000000000	0x000000000
0x10010020	0x000000000	0x000000000	0x000000000	0x000000000	0x000000000	0x000000000	0x000000000	0x000000000
0x10010040	0x000000000	0x000000000	0x000000000	0x000000000	0x000000000	0x000000000	0x000000000	0x000000000
0x10010060	0x000000000	0x000000000	0x000000000	0x000000000	0x000000000	0x000000000	0x000000000	0x000000000
0x10010080	0x000000000	0x000000000	0x000000000	0x000000000	0x000000000	0x000000000	0x000000000	0x000000000
0x100100A0	0x000000000	0x000000000	0x000000000	0x000000000	0x000000000	0x000000000	0x000000000	0x000000000
0x100100C0	0x000000000	0x000000000	0x000000000	0x000000000	0x000000000	0x000000000	0x000000000	0x000000000
0x100100E0	0x000000000	0x000000000	0x000000000	0x000000000	0x000000000	0x000000000	0x000000000	0x000000000
0x10010100	0x000000000	0x000000000	0x000000000	0x000000000	0x000000000	0x000000000	0x000000000	0x000000000
0x10010120	0x000000000	0x000000000	0x000000000	0x000000000	0x000000000	0x000000000	0x000000000	0x000000000
0x10010140	0x000000000	0x000000000	0x000000000	0x000000000	0x000000000	0x000000000	0x000000000	0x000000000
0x10010160	0x000000000	0x000000000	0x000000000	0x000000000	0x000000000	0x000000000	0x000000000	0x000000000
0x10010180	0x000000000	0x000000000	0x000000000	0x000000000	0x000000000	0x000000000	0x000000000	0x000000000
0x100101A0	0x000000000	0x000000000	0x000000000	0x000000000	0x000000000	0x000000000	0x000000000	0x000000000

Edit ☒ Hexadecimal Addresses ☒ Hexadecimal Values ☐ ASCII

Programa 14 –

```

p8.asm p9.asm p5.asm p10.asm p11.asm p12.asm p13.asm p14.asm
1  #programa 14
2
3  .data
4  A: .word 4
5
6  .text
7  ori $t0, $zero, 0x1001
8  sll $t0, $t0, 16
9
10 lv $t1, 0($t0)
11
12 ori $t2, $zero, 1
13
14 and $t3, $t1, $t2 #o resultado do and da 0 quando e par e 1 quando e impar
15 # (pois quando e impar, o bit menos significativo tem um 1, e fznd and com ele, é o unico que retornara 1, diferente do 0 final dos numeros pares)
16 # 0001 and 0001 -> 0001
17 # 0010 and 0001 -> 0000
18
19 sw $t3, 4($t0) #escreve o valor na segunda posicao livre da memoria (0x10010004)

```

Edit Execute

Text Segment

Byte	Address	Code	Basic	Source
0x00400000	0x34081001	ori \$t0, \$zero, 0x1001	7: ori \$t0, \$zero, 0x1001	
0x00400004	0x00084400	sll \$t0, \$t0, 16	8: sll \$t0, \$t0, 16	
0x00400008	0x00089000	lv \$t1, 0(\$t0)	10: lv \$t1, 0(\$t0)	
0x0040000c	0x340a0001	ori \$t2, \$zero, 1	12: ori \$t2, \$zero, 1	
0x00400010	0x012a5824	and \$t3, \$t1, \$t2	14: and \$t3, \$t1, \$t2 #o resultado do and da 0 quando e par e 1 quando e impar	
0x00400014	0x000b0004	sw \$t3, 4(\$t0)	19: sw \$t3, 4(\$t0) #escreve o valor na segunda posicao livre da memoria (0x10010004)	

Data Segment

Address	Value (+0)	Value (+4)	Value (+8)	Value (+c)	Value (+10)	Value (+14)	Value (+18)	Value (+1c)
0x10010000	0x00000004	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x10010020	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x10010040	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x10010060	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x10010080	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x100100a0	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x100100c0	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x100100e0	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x10010100	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x10010120	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x10010140	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x10010160	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x10010180	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x100101a0	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000

Registers Coproc 1 Coproc 0

Name	Number	Value
\$zero	0	0x00000000
\$at	1	0x00000000
\$v0	2	0x00000000
\$v1	3	0x00000000
\$a0	4	0x00000000
\$a1	5	0x00000000
\$a2	6	0x00000000
\$a3	7	0x00000000
\$t0	8	0x10010001
\$t1	9	0x00000004
\$t2	10	0x00000001
\$t3	11	0x00000000
\$t4	12	0x00000000
\$t5	13	0x00000000
\$t6	14	0x00000000
\$t7	15	0x00000000
\$t8	16	0x00000000
\$t9	17	0x00000000
\$s0	18	0x00000000
\$s1	19	0x00000000
\$s2	20	0x00000000
\$s3	21	0x00000000
\$s4	22	0x00000000
\$s5	23	0x00000000
\$s6	24	0x00000000
\$s7	25	0x00000000
\$s8	26	0x00000000
\$s9	27	0x00000000
\$gp	28	0x10000000
\$sp	29	0x7fffffc0
\$fp	30	0x00000000
\$ra	31	0x00000000
\$pc		0x00400018
\$hi		0x00000000
\$lo		0x00000000

Programa 15 –

```

1  .data
2  vetor: .space 400 # alocando espaço para 100 inteiros (4 bytes cada)
3  soma: .word 0
4
5  .text
6  main:
7      la $t0, vetor      # endereço base
8      li $t1, 0          # (índice)
9      li $t2, 0          # (soma)
10
11 loop_inicio:
12     bge $t1, 100, loop_fim # se $t1 >= 100, sai do loop de inicio
13
14     # calcula o vetor[i] = 2*i + 1
15     sll $t4, $t1, 1      # $t4 = 2 * $t1 (i * 2)
16     addi $t4, $t4, 1     # $t4 = (i * 2) + 1
17
18     # armazena o vetor[i]
19     sll $t5, $t1, 2      # $t5 = i * 4 (índice em bytes)
20     add $t6, $t0, $t5    # $t6 = endereço de vetor[i]
21     sw $t4, 0($t6)      # vetor[i] = $t4
22
23     add $t2, $t2, $t4    # atualiza a soma
24
25     addi $t1, $t1, 1     # i++
26     j loop_inicio       # repete o loop
27
28 loop_fim:
29
30     la $t6, soma         # carrega o endereço de soma em $t6
31     sw $t2, 0($t6)      # armazena a soma

```

Assembly code editor showing assembly instructions and registers.

Text Segment

Bit	Address	Code	Basic	Source
	0x00400000	0x3c011001	lui \$t0, 0x0001001	7: la \$t0, vetor # endereço base
	0x00400004	0x34280000	ori \$t1, 0x00000000	
	0x00400008	0x40900000	addiu \$t5, \$t0, 0x00000000	8: li \$t5, 0 # (índice)
	0x0040000c	0x24000000	addiu \$t0, \$t0, 0x00000000	9: li \$t2, 0 # (soma)
	0x00400010	0x20210004	blez \$t1, \$t5, 0x00000004	12: bge \$t1, 100, loop_fim # se \$t1 >= 100, sai do loop de início
	0x00400014	0x10200000	beq \$t1, \$t5, 0x00000000	
	0x00400018	0x00900000	sll \$t4, \$t1, 1	15: sll \$t4, \$t1, 1 # \$t4 = 2 * \$t1 (1 * 2)
	0x0040001c	0x18000001	addi \$t4, \$t4, 1	16: addi \$t4, \$t4, 1 # \$t4 = (1 * 2) + 1
	0x00400020	0x00000000	sll \$t5, \$t1, 2	19: sll \$t5, \$t1, 2 # \$t5 = 1 * 4 (índice em bytes)
	0x00400024	0x10070020	add \$t6, \$t0, \$t5	20: add \$t6, \$t0, \$t5 # \$t6 = endereço de vetor[i]
	0x00400028	0x00000000	sw \$t4, 0(\$t6)	21: sw \$t4, 0(\$t6) # vetor[i] = \$t4
	0x0040002c	0x014c0020	add \$t0, \$t0, \$t2	23: add \$t0, \$t0, \$t2 # atualiza a soma
	0x00400030	0x12000001	addi \$t1, \$t1, 1	25: addi \$t1, \$t1, 1 # i++
	0x00400034	0x00100040	Op00400040	26: j loop_inicio # repete o loop
	0x00400038	0x3c011001	lui \$t0, 0x0001001	30: la \$t6, soma # carrega o endereço de soma em \$t6
	0x0040003c	0x342e0190	ori \$t1, \$t1, 0x00000190	
	0x00400040	0x00000000	Op00000000	31: Op00000000 # apresenta a soma

Data Segment

Address	Value (+0)	Value (+4)	Value (+8)	Value (+c)	Value (+10)	Value (+14)	Value (+18)	Value (+1c)
0x10010000	0x00000001	0x00000003	0x00000005	0x00000007	0x00000009	0x0000000b	0x0000000d	0x0000000f
0x10010004	0x00000011	0x00000013	0x00000015	0x00000017	0x00000019	0x0000001b	0x0000001d	0x0000001f
0x10010008	0x00000021	0x00000023	0x00000025	0x00000027	0x00000029	0x0000002b	0x0000002d	0x0000002f
0x1001000c	0x00000031	0x00000033	0x00000035	0x00000037	0x00000039	0x0000003b	0x0000003d	0x0000003f
0x10010010	0x00000041	0x00000043	0x00000045	0x00000047	0x00000049	0x0000004b	0x0000004d	0x0000004f
0x10010014	0x00000051	0x00000053	0x00000055	0x00000057	0x00000059	0x0000005b	0x0000005d	0x0000005f
0x10010018	0x00000061	0x00000063	0x00000065	0x00000067	0x00000069	0x0000006b	0x0000006d	0x0000006f
0x1001001c	0x00000071	0x00000073	0x00000075	0x00000077	0x00000079	0x0000007b	0x0000007d	0x0000007f
0x10010020	0x00000081	0x00000083	0x00000085	0x00000087	0x00000089	0x0000008b	0x0000008d	0x0000008f
0x10010024	0x00000091	0x00000093	0x00000095	0x00000097	0x00000099	0x0000009b	0x0000009d	0x0000009f
0x10010028	0x000000a1	0x000000a3	0x000000a5	0x000000a7	0x000000a9	0x000000ab	0x000000ad	0x000000af
0x1001002c	0x000000b1	0x000000b3	0x000000b5	0x000000b7	0x000000b9	0x000000bb	0x000000bd	0x000000bf
0x10010030	0x000000c1	0x000000c3	0x000000c5	0x000000c7	0x000000c9	0x000000cb	0x000000cd	0x000000cf
0x10010034	0x000000d1	0x000000d3	0x000000d5	0x000000d7	0x000000d9	0x000000db	0x000000dd	0x000000df
0x10010038	0x000000e1	0x000000e3	0x000000e5	0x000000e7	0x000000e9	0x000000eb	0x000000ed	0x000000ef
0x1001003c	0x000000f1	0x000000f3	0x000000f5	0x000000f7	0x000000f9	0x000000fb	0x000000fd	0x000000ff

Registers

Name	Number	Value
\$zero	0	0x00000000
\$at	1	0x00000000
\$v0	2	0x00000000
\$v1	3	0x00000000
\$a0	4	0x00000000
\$a1	5	0x00000000
\$a2	6	0x00000000
\$a3	7	0x00000000
\$t0	8	0x00000000
\$t1	9	0x00000000
\$t2	10	0x00000000
\$t3	11	0x00000000
\$t4	12	0x00000000
\$t5	13	0x00000000
\$t6	14	0x00000000
\$t7	15	0x00000000
\$s0	16	0x00000000
\$s1	17	0x00000000
\$s2	18	0x00000000
\$s3	19	0x00000000
\$s4	20	0x00000000
\$s5	21	0x00000000
\$s6	22	0x00000000
\$s7	23	0x00000000
\$t9	24	0x00000000
\$t10	25	0x00000000
\$t11	26	0x00000000
\$t12	27	0x00000000
\$fp	28	0x00000000
\$gp	29	0x00000000
\$ra	30	0x00000000
\$lr	31	0x00000000
\$pc		0x00000000
\$hi		0x00000000
\$lo		0x00000000

Programa 16 –

```
1  #programa 16
2  .data
3  x:.word 0x186A00 # x = 160000
4  y:.word 0x13880 # y = 8000
5  z:.word 0x61A80 # z = 40000
6
7  .text
8  lui $t0, 0x1001
9  lw $s0, 0($t0) #s0 = x
10 lw $s1, 4($t0) #s1 = y
11 lw $s2, 8($t0) #s2 = z
12
13 mult $s0, $s1 # x * y
14 mflo $t1 # resultado da multiplicacao
15
16 div $t1, $s2
17 mflo $t2 # quociente da divisao de (x * y)/z
```

Assembly code editor showing assembly instructions and registers.

Text Segment

Bit	Address	Code	Basic	Source
	0x00400000	0x3c011001	lui \$t0, 0x0001001	8: lui \$t0, 0x1001
	0x00400004	0x8d100000	lw \$s0, 0(\$t0)	9: lw \$s0, 0(\$t0) #s0 = x
	0x00400008	0x8d110004	lw \$s1, 4(\$t0)	10: lw \$s1, 4(\$t0) #s1 = y
	0x0040000c	0x8d120008	lw \$s2, 8(\$t0)	11: lw \$s2, 8(\$t0) #s2 = z
	0x00400010	0x00210018	mult \$s0, \$s1	13: mult \$s0, \$s1 # x * y
	0x00400014	0x00004112	mflo \$t1	14: mflo \$t1 # resultado da multiplicacao
	0x00400018	0x00130014	div \$t1, \$s2	16: div \$t1, \$s2
	0x0040001c	0x00005012	mflo \$t2	17: mflo \$t2 # quociente da divisao de (x * y)/z

Data Segment

Address	Value (+0)	Value (+4)	Value (+8)	Value (+c)	Value (+10)	Value (+14)	Value (+18)	Value (+1c)
0x10010000	0x00186a00	0x00013880	0x00061a80	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x10010004	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x10010008	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x1001000c	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x10010010	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x10010014	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x10010018	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x1001001c	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x10010020	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x10010024	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x10010028	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x1001002c	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x10010030	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x10010034	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x10010038	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x1001003c	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000

Registers

Name	Number	Value
\$zero	0	0x00000000
\$at	1	0x00000000
\$v0	2	0x00000000
\$v1	3	0x00000000
\$a0	4	0x00000000
\$a1	5	0x00000000
\$a2	6	0x00000000
\$a3	7	0x00000000
\$t0	8	0x00000000
\$t1	9	0x00000000
\$t2	10	0xffffffff
\$t3	11	0x00000000
\$t4	12	0x00000000
\$t5	13	0x00000000
\$t6	14	0x00000000
\$t7	15	0x00000000
\$s0	16	0x00186a00
\$s1	17	0x00061a80
\$s2	18	0x00000000
\$s3	19	0x00000000
\$s4	20	0x00000000
\$s5	21	0x00000000
\$s6	22	0x00000000
\$s7	23	0x00000000
\$t9	24	0x00000000
\$t10	25	0x00000000
\$t11	26	0x00000000
\$t12	27	0x00000000
\$fp	28	0x00000000
\$gp	29	0xffffffff
\$ra	30	0x00000000
\$lr	31	0x00000000
\$pc		0x00400020
\$hi		0xffffffff
\$lo		0xffffffff

Programa 17–

```

1  #programa 17
2  .data
3  x: .word 5
4  y: .word 10
5
6  .text
7  lui $t0, 0x1001
8  lw $t1, 0($t0) # x = 5
9  lw $t2, 4($t0) # y = 10 (e vai servir de indice no loop)
10 ori $t3, $zero, 1
11
12 loop_inicio:
13 beq $t2, $zero, loop_fim # se o indice chegar a 0, vai para loop_fim
14 sub $t2, $t2, $t3 # decrementa o indice
15 add $t4, $t4, $t1 #itera a multiplicacao
16 j loop_inicio
17
18 loop_fim:
19 sw $t4, 8($t0)

```

Text Segment

Bkpt

Address

Code

Basic

Source

0x00400000

0x30081001

lui \$5, 0x000001001

7: lui \$t0, 0x1001

0x00400004

0x80900000

lw \$9, 0x00000000(\$t0)

8: lw \$t1, 0(\$t0) # x = 5

0x00400008

0x80400004

lw \$10, 0x00000004(\$t0)

9: lw \$t2, 4(\$t0) # y = 10 (e vai servir de indice no loop)

0x0040000c

0x340b0001

ori \$t1, \$zero, 1

10: ori \$t3, \$zero, 1

0x00400010

0x11400003

beq \$t2, \$zero, 0x00000003

13: beq \$t2, \$zero, loop_fim # se o indice chegar a 0, vai para loop_fim

0x00400014

0x04050022

sub \$t0, \$t0, \$t1

14: sub \$t2, \$t2, \$t3 # decrementa o indice

0x00400018

0x01800020

add \$t2, \$t2, \$t3

15: add \$t4, \$t4, \$t3 # incrementa o indice

0x0040001c

0x08100004

j 0x00400010

16: j loop_inicio

0x00400020

0x0ad0c008

sw \$t2, 0x00000008(\$t0)

19: sw \$t4, 8(\$t0)

Name

Number

Value

\$zero

0

0x00000000

\$at

1

0x00000000

\$v0

2

0x00000000

\$v1

3

0x00000000

\$a0

4

0x00000000

\$a1

5

0x00000000

\$a2

6

0x00000000

\$a3

7

0x00000000

\$t0

8

0x00100000

\$t1

9

0x00000005

\$t2

10

0x00000000

\$t3

11

0x00000001

\$t4

12

0x00000032

\$t5

13

0x00000000

\$t6

14

0x00000000

\$t7

15

0x00000000

\$t8

16

0x00000000

\$t9

17

0x00000000

\$a2

18

0x00000000

\$a3

19

0x00000000

\$t4

20

0x00000000

\$t5

21

0x00000000

\$t6

22

0x00000000

\$a7

23

0x00000000

\$a8

24

0x00000000

\$t9

25

0x00000000

\$t0

26

0x00000000

\$t1

27

0x00000000

\$t2

28

0x00000000

\$t3

29

0x00000000

\$t4

30

0x00000000

\$t5

31

0x00000000

\$t6

32

0x00000000

\$t7

33

0x00000000

\$t8

34

0x00000000

\$t9

35

0x00000000

\$a0

36

0x00000000

\$a1

37

0x00000000

\$a2

38

0x00000000

\$a3

39

0x00000000

\$a4

40

0x00000000

\$a5

41

0x00000000

\$a6

42

0x00000000

\$a7

43

0x00000000

\$a8

44

0x00000000

\$a9

45

0x00000000

\$a10

46

0x00000000

\$a11

47

0x00000000

\$a12

48

0x00000000

\$a13

49

0x00000000

\$a14

50

0x00000000

\$a15

51

0x00000000

\$a16

52

0x00000000

\$a17

53

0x00000000

\$a18

54

0x00000000

\$a19

55

0x00000000

\$a20

56

0x00000000

\$a21

57

0x00000000

\$a22

58

0x00000000

\$a23

59

0x00000000

\$a24

60

0x00000000

\$a25

61

0x00000000

Data Segment

Address

Value (+0)

Value (+4)

Value (+8)

Value (+C)

Value (+10)

Value (+14)

Value (+18)

Value (+1C)

0x10010000

0x00000005

0x0000000A

0x00000032

0x00000000

0x00000000

0x00000000

0x00000000

0x10010020

0x00000000

0x00000000

0x00000000

0x00000000

0x00000000

0x00000000

0x00000000

0x10010040

0x00000000

0x00000000

0x00000000

0x00000000

0x00000000

0x00000000

0x00000000

0x10010060

0x00000000

0x00000000

0x00000000

0x00000000

0x00000000

0x00000000

0x00000000

0x10010080

0x00000000

0x00000000

0x00000000

0x00000000

0x00000000

0x00000000

0x00000000

0x100100A0

0x00000000

0x00000000

0x00000000

0x00000000

0x00000000

0x00000000

0x00000000

0x100100C0

0x00000000

0x00000000

0x00000000

0x00000000

0x00000000

0x00000000

0x00000000

0x100100E0

0x00000000

0x00000000

0x00000000

0x00000000

0x00000000

0x00000000

0x00000000

0x10010100

0x00000000

0x00000000

0x00000000

0x00000000

0x00000000

0x00000000

0x00000000

0x10010120

0x00000000

0x00000000

0x00000000

0x00000000

0x00000000

0x00000000

0x00000000

0x10010140

0x00000000

0x00000000

0x00000000

0x00000000

0x00000000

0x00000000

0x00000000

0x10010160

0x00000000

0x00000000

0x00000000

0x00000000

0x00000000

0x00000000

0x00000000

0x10010180

0x00000000

0x00000000

0x00000000

0x00000000

0x00000000

0x00000000

0x00000000

0x100101A0

0x00000000

0x00000000

0x00000000

0x00000000

0x00000000

0x00000000

0x00000000

Programa 18–

```

1  #programa 18
2  #k = x ^ y
3  .data
4  x: .word 5
5  y: .word 3
6
7  .text
8  lui $t0, 0x1001
9  lw $t1, 0($t0) # x = 5
10 lw $t2, 4($t0) # y = 3 (e vai servir de indice no loop)
11 ori $t3, $zero, 1
12 ori $t5, $zero, 5
13
14 loop_inicio:
15 beq $t2, 1, loop_fim # se o indice chegar a 1, vai para loop_fim
16 mult $t1, $t5
17 mflo $t5 #resultado de x * x
18 add $t5, $t5, $t4 # adiciona (x*x) ao resultado da elevacao final
19 sub $t2, $t2, $t3 # decrementa o indice
20 j loop_inicio
21
22 loop_fim:
23 sw $t4, 8($t0)

```

Text Segment				Name		Number	Value
Bitset	Address	Code	Basic	Source			
☐	0x00400000	0x00001001	lui \$t0, 0x00001001	8: lui \$t0, 0x00001001	\$zero	0	0x00000000
☐	0x00400004	0x0d090000	lw \$t9, 0x00000000(\$t0)	9: lw \$t1, 0(\$t0) # x = 5	\$at	1	0x00000001
☐	0x00400008	0x0d0a0004	lw \$t0, 0x00000004(\$t0)	10: lw \$t2, 4(\$t0) # y = 3 (e vai servir de indice no loop)	\$v1	2	0x00000000
☐	0x0040000c	0x340b0001	ori \$t0, \$zero, 1	11: ori \$t2, \$zero, 1	\$a0	3	0x00000000
☐	0x00400010	0x0d0a0005	ori \$t1, \$t0, 0x00000005	12: ori \$t5, \$zero, 5	\$a1	4	0x00000000
☐	0x00400014	0x02010001	addi \$t1, \$t0, 0x00000001	15: beq \$t2, 1, loop_fim # se o indice chegar a 1, vai para loop_fim	\$a2	5	0x00000000
☐	0x00400018	0x102a0005	beq \$t1, \$t0, 0x00000005		\$a3	6	0x00000000
☐	0x0040001c	0x012a0011	mul \$t1, \$t5, \$t13	14: mul \$t1, \$t5	\$t0	8	0x00100000
☐	0x00400020	0x00006812	mflo \$t3	17: mflo \$t5 #resultado de x * x	\$t1	9	0x00000005
☐	0x00400024	0x01ac0020	add \$t3, \$t5, \$t12	18: add \$t5, \$t5, \$t2 # adiciona (x*x) ao resultado da elevacao final	\$t2	10	0x00000001
☐	0x00400028	0x012a0011	mul \$t1, \$t5, \$t13	19: mul \$t2, \$t2, \$t3 # decrementa o indice	\$t3	11	0x00000001
☐	0x0040002c	0x05100005	0x04000014	20: j loop_inicio	\$t4	12	0x00000000
☐	0x00400030	0x0ad00008	sw \$t1, 0x00000008(\$t0)	23: sw \$t4, 8(\$t0)	\$t5	13	0x00000000
☐					\$t6	14	0x00000000
☐					\$t7	15	0x00000000
☐					\$t8	16	0x00000000
☐					\$t9	17	0x00000000
☐					\$t10	18	0x00000000
☐					\$t11	19	0x00000000
☐					\$t12	20	0x00000000
☐					\$t13	21	0x00000000
☐					\$t14	22	0x00000000
☐	0x00100000	0x00000005	0x00000003	0x00000000	\$t15	23	0x00000000
☐	0x00100020	0x00000000	0x00000000	0x00000000	\$t16	24	0x00000000
☐	0x00100040	0x00000000	0x00000000	0x00000000	\$t17	25	0x00000000
☐	0x00100060	0x00000000	0x00000000	0x00000000	\$t18	26	0x00000000
☐	0x00100080	0x00000000	0x00000000	0x00000000	\$t19	27	0x00000000
☐	0x001000a0	0x00000000	0x00000000	0x00000000	\$t20	28	0x00000000
☐	0x001000c0	0x00000000	0x00000000	0x00000000	\$t21	29	0x00000000
☐	0x001000e0	0x00000000	0x00000000	0x00000000	\$t22	30	0x00000000
☐	0x00100100	0x00000000	0x00000000	0x00000000	\$t23	31	0x00000000
☐	0x00100120	0x00000000	0x00000000	0x00000000	\$t24	32	0x00000000
☐	0x00100140	0x00000000	0x00000000	0x00000000	\$t25	33	0x00000000
☐	0x00100160	0x00000000	0x00000000	0x00000000	\$t26	34	0x00000000
☐	0x00100180	0x00000000	0x00000000	0x00000000	\$t27	35	0x00000000
☐	0x001001a0	0x00000000	0x00000000	0x00000000	\$t28	36	0x00000000
☐					\$t29	37	0x00000000
☐					\$t30	38	0x00000000
☐					\$t31	39	0x00000000
☐					\$t32	40	0x00000000
☐					\$t33	41	0x00000000
☐					\$t34	42	0x00000000
☐					\$t35	43	0x00000000
☐					\$t36	44	0x00000000
☐					\$t37	45	0x00000000
☐					\$t38	46	0x00000000
☐					\$t39	47	0x00000000
☐					\$t40	48	0x00000000
☐					\$t41	49	0x00000000
☐					\$t42	50	0x00000000
☐					\$t43	51	0x00000000
☐					\$t44	52	0x00000000
☐					\$t45	53	0x00000000
☐					\$t46	54	0x00000000
☐					\$t47	55	0x00000000
☐					\$t48	56	0x00000000
☐					\$t49	57	0x00000000
☐					\$t50	58	0x00000000
☐					\$t51	59	0x00000000
☐					\$t52	60	0x00000000
☐					\$t53	61	0x00000000
☐					\$t54	62	0x00000000
☐					\$t55	63	0x00000000
☐					\$t56	64	0x00000000
☐					\$t57	65	0x00000000
☐					\$t58	66	0x00000000
☐					\$t59	67	0x00000000
☐					\$t60	68	0x00000000
☐					\$t61	69	0x00000000
☐					\$t62	70	0x00000000
☐					\$t63	71	0x00000000
☐					\$t64	72	0x00000000
☐					\$t65	73	0x00000000
☐					\$t66	74	0x00000000
☐					\$t67	75	0x00000000
☐					\$t68	76	0x00000000
☐					\$t69	77	0x00000000
☐					\$t70	78	0x00000000
☐					\$t71	79	0x00000000
☐					\$t72	80	0x00000000
☐					\$t73	81	0x00000000
☐					\$t74	82	0x00000000
☐					\$t75	83	0x00000000
☐					\$t76	84	0x00000000
☐					\$t77	85	0x00000000
☐					\$t78	86	0x00000000
☐					\$t79	87	0x00000000
☐					\$t80	88	0x00000000
☐					\$t81	89	0x00000000
☐					\$t82	90	0x00000000
☐					\$t83	91	0x00000000
☐					\$t84	92	0x00000000
☐					\$t85	93	0x00000000
☐					\$t86	94	0x00000000
☐					\$t87	95	0x00000000
☐					\$t88	96	0x00000000
☐					\$t89	97	0x00000000
☐					\$t90	98	0x00000000
☐					\$t91	99	0x00000000
☐					\$t92	100	0x00000000
☐					\$t93	101	0x00000000
☐					\$t94	102	0x00000000
☐					\$t95	103	0x00000000
☐					\$t96	104	0x00000000
☐					\$t97	105	0x00000000
☐					\$t98	106	0x00000000
☐					\$t99	107	0x00000000
☐					\$t100	108	0x00000000
☐					\$t101	109	0x00000000
☐					\$t102	110	0x00000000
☐					\$t103	111	0x00000000
☐					\$t104	112	0x00000000
☐					\$t105	113	0x00000000
☐					\$t106	114	0x00000000
☐					\$t107	115	0x00000000
☐					\$t108	116	0x00000000
☐					\$t109	117	0x00000000
☐					\$t110	118	0x00000000
☐					\$t111	119	0x00000000
☐					\$t112	120	0x00000000
☐					\$t113	121	0x00000000
☐					\$t114	122	0x00000000
☐					\$t115	123	0x00000000
☐					\$t116	124	0x00000000
☐					\$t117	125	0x00000000
☐					\$t118	126	0x00000000
☐					\$t119	127	0x00000000
☐					\$t120	128	0x00000000
☐					\$t121	129	0x00000000
☐					\$t122	130	0x00000000
☐					\$t123	131	0x00000000
☐					\$t124	132	0x00000000
☐					\$t125	133	0x00000000
☐					\$t126	134	0x00000000
☐					\$t127	135	0x00000000
☐					\$t128	136	0x00000000
☐					\$t129	137	0x00000000
☐					\$t130	138	0x00000000
☐					\$t131	139	0x00000000
☐					\$t132	140	0x00000000
☐					\$t133	141	0x00000000
☐					\$t134	142	0x00000000
☐					\$t135	143	0x00000000
☐					\$t136	144	0x00000000
☐					\$t137	145	0x00000000
☐					\$t138	146	0x00000000
☐					\$t139	147	0x00000000
☐					\$t140	148	0x00000000
☐					\$t141	149	0x00000000
☐					\$t142	150	0x00000000
☐					\$t143	151	0x00000000
☐					\$t144	152	0x00000000
☐					\$t145	153	0x00000000
☐					\$t146	154	0x00000000
☐					\$t147	155	0x00000000
☐					\$t148	156	0x00000000
☐					\$t149	157	0x00000000
☐					\$t150	158	0x00000000
☐					\$t151	159	0x00000000
☐					\$t152	160	0x00000000
☐					\$t153	161	0x00000000
☐					\$t154	162	0x00000000
☐					\$t155	163	0x00000000
☐					\$t156	164	0x00000000
☐					\$t157	165	0x00000000
☐					\$t158	166	0x00000000
☐					\$t159	167	0x00000000
☐					\$t160	168	0x00000000
☐					\$t161	169	0x00000000
☐					\$t162	170	0x00000000
☐					\$t163	171	0x00000000
☐					\$t164	172	0x00000000
☐					\$t165	173	0x00000000
☐					\$t166	174	0x00000000
☐					\$t167	175	0x00000000
☐					\$t168	176	0x00000000
☐					\$t169	177	0x00000000
☐							

Parte 3 – Respostas

1. Se tivermos 2 inteiros, cada um com 32 bits, quantos bits podemos esperar para o produto?

Resp: Letra C - 32

2. Quais os registradores que armazenam os resultados na multiplicação?

Resp: Letra B - hi e lo

3. Qual a operação usada para multiplicar inteiros em comp. de dois?

Resp: Letra A - mult

4. Qual instrução move os bits menos significativos da multiplicação para o reg. 8?

Resp: Letra C - mflo \$8

5. Se tivermos dois inteiros, cada um com 32 bits, quantos bits deveremos estar preparados para receber no quociente?

Resp: Letra B - 32

6. Após a instrução div, qual registrador possui o quociente?

Resp: Letra A - lo

7. Qual a inst. Usada para dividir dois inteiros em comp. de dois?

Resp: Letra D - div

8. Faça um arithmetic shift right de dois no seguinte padrão de bits: 1001 1011

Resp: Letra B - 0010 0110

9. Qual o efeito de um arithmetic shift right de uma posição?

Resp: Letra A - Se o inteiro for unsigned, o shift o divide por 2. Se o inteiro for signed, o shift o divide por 2.

10. Qual sequencia de instruções avalia $3x+7$, onde x é iniciado no reg. \$8 e o resultado armazenado em \$9?

Resp: Letra A - ori \$3,\$0,3 / mult \$8,\$3 / mflo \$9 / addi \$9,\$9,7

Parte 4 – Programas MIPS/MARS

Programa 19–

```
p1.asm  p8.asm  p5.asm  p10.asm  p11.asm  p12.asm  p13.asm  p14.asm  p15.asm  p16.asm  p17.asm  p18.asm  p19.asm
1  #programa 19
2  .data
3  x: .word 0xFFFF
4  y: .word 0xFFFF
5  .text
6  lui $t0, 0x1001
7  lw $a0, 0($t0)
8  lw $a1, 4($t0)
9  ori $t6, $zero, 32 # $t6 -> valor 32
10 ori $t7, $zero, 1
11 or $t1, $zero, $a0 # copia o valor de $a0 para $t1
12 ori $t2, $zero, 31
13 or $t4, $zero, $t1
14 loop:
15 beq $t5, $zero, repetir # Se $t5 não for zero, vai para repetir
16 ori $t3, $t4, 31 # isole o bit mais significativo
17 beq $t3, $t7, repetir # Se $t5 for igual a 1, vai para repetir
18 sll $t4, $t4, 1 # Desloca $t4 1 bit para a esquerda
19 addi $t2, $t2, -1 # $t6--
20 j loop
21
22 repetir:
23 beq $t1, $t1, final_qnt_bits # Se $t1 for igual a $t1, vai para final_qnt_bits
24 or $t0, $zero, $t6
25 or $t1, $zero, $t1
26 or $t4, $zero, $t1
27 or $t5, $zero, $zero
28 ori $t6, $zero, 32
29 j loop
30
31 final_qnt_bits:
32 or $t1, $zero, $t6
33 ori $t6, $zero, 32
34 mult $t0, $t1
35 add $t4, $t0, $t1
36 sll $t4, $t4, $t6 # Define $t4 como 1 se $t4 for menor que $t6, se não define como 0
37 beq $t4, $zero, maior_32 # Se $t4 for igual a zero, vai para maior_32
38 mflo $a2
39 j fim
40
41 maior_32:
42 mfli $a2
43 mfli $a3
44 fim
```

Assembly code editor showing MIPS instructions and registers. The main window displays assembly code with comments in Portuguese. The registers window on the right shows the state of registers \$zero through \$31.

Registers	Coproc 1	Coproc 0	Name	Number	Value
			\$zero	0	0x00000000
			\$at	1	0x00000000
			\$v0	2	0x00000000
			\$v1	3	0x00000000
			\$a0	4	0x00000000
			\$a1	5	0x00000000
			\$a2	6	0x00000000
			\$a3	7	0x00000000
			\$t0	8	0x10010000
			\$t1	9	0x00000002
			\$t2	10	0x0000000f
			\$t3	11	0x00000000
			\$t4	12	0x00000000
			\$t5	13	0x00000001
			\$t6	14	0x00000020
			\$t7	15	0x00000001
			\$d0	16	0x000000ff
			\$d1	17	0x000000ff
			\$d2	18	0x000000ff
			\$d3	19	0x00000001
			\$d4	20	0x00000000
			\$d5	21	0x00000000
			\$d6	22	0x00000000
			\$d7	23	0x00000000
			\$t8	24	0x00000000
			\$t9	25	0x00000000
			\$k0	26	0x00000000
			\$k1	27	0x00000000
			\$k2	28	0x00000000
			\$k3	29	0xfffffff
			\$k4	30	0x00000000
			\$k5	31	0x00000000
			\$pc		0x0040007c
			\$hi		0x000000ff
			\$lo		0xffff0001

Programa 20–

```
1 #programa 20
2 .data
3 x: word 3
4
5 .text
6 lui $t0, 0x1001
7 ori $t1, $zero, 2
8 lw $s0, 0($t0)
9 div $s0, $t1
10 mfhi $s1
11 mult $s0, $s0
12 mflo $s2 # resultado de x^2
13 mult $s2, $s0
14 mflo $s3 # resultado de x^3
15 mult $s3, $s0
16 mflo $s4 # resultado de x^4
17 mult $s4, $s0
18 mflo $s5 # resultado de x^5
19
20 beq $s1, $zero, par
21 j impar
22
23 par:
24 add $t2, $s4, $s3
25 mult $t1, $s2
26 mflo $s6 # 2*x^2
27 sub $t3, $t2, $s6
28 sw $t3, 4($t0)
29 j fim
30
31 impar:
32 sub $t4, $s5, $s3
33 addi $s7, $t4, 1
34 sw $s7, 4($t0)
35
36 fim:
```

Assembly code editor showing MIPS instructions and registers. The main window displays assembly code with comments in Portuguese. The registers window on the right shows the state of registers \$zero through \$31.

Registers	Coproc 1	Coproc 0	Name	Number	Value
			\$zero	0	0x00000000
			\$at	1	0x00000000
			\$v0	2	0x00000000
			\$v1	3	0x00000000
			\$a0	4	0x00000000
			\$a1	5	0x00000000
			\$a2	6	0x00000000
			\$a3	7	0x00000000
			\$t0	8	0x10010000
			\$t1	9	0x00000002
			\$t2	10	0x0000000f
			\$t3	11	0x00000000
			\$t4	12	0x00000000
			\$t5	13	0x00000000
			\$t6	14	0x00000000
			\$t7	15	0x00000000
			\$d0	16	0x00000000
			\$d1	17	0x00000001
			\$d2	18	0x00000000
			\$d3	19	0x00000000
			\$d4	20	0x00000051
			\$d5	21	0x000000f3
			\$d6	22	0x00000000
			\$d7	23	0x00000000
			\$t8	24	0x00000000
			\$t9	25	0x00000000
			\$k0	26	0x00000000
			\$k1	27	0x00000000
			\$k2	28	0x00000000
			\$k3	29	0xfffffff
			\$k4	30	0x00000000
			\$k5	31	0x00000000
			\$pc		0x0040007c
			\$hi		0x000000ff
			\$lo		0x000000f3

Programa 21–

```

1  #programa 21
2  .data
3  x: .word 2
4
5  .text
6  lui $t0, 0x1001
7  lw $s0, 0($t0) # x
8  mult $s0, $s0
9  mflo $s2 # resultado de x^2
10 mult $s2, $s0
11 mflo $s3 # resultado de x^3
12 mult $s3, $s0
13 mflo $s4 # resultado de x^4
14 mult $s4, $s0
15 mflo $s5 # resultado de x^5
16
17 slt $t1, $s0, $zero #set less than -> se s0 for menor que zero, seta t1 como 1, se nao, 0.
18 beq $t1, $zero, maior
19 j menor
20
21 maior:
22 addi $t2, $s3, 1
23 sw $t2, 4($t0)
24 j fim
25
26 menor:
27 addi $t3, $s4, 0xFFFFFFFF
28 sw $t3, 4($t0)
29
30 fim:

```

Text Segment

Bkpt	Address	Code	Basic	Source
☐	0x00400000	0x00000000	lui \$5, 0x00000001	6: lui \$5, 0x00000001
☐	0x00400004	0x00100001	lw \$16, 0x00000000(\$5)	7: lw \$a0, 0(\$t0) * \$
☐	0x00400008	0x00100001	mult \$16, \$16	8: mult \$a0, \$a0
☐	0x0040000c	0x00000000	mflo \$18	9: mflo \$a2 * resultado de x*3
☐	0x00400010	0x00200001	mult \$18, \$16	10: mult \$a2, \$a0
☐	0x00400014	0x00000002	mflo \$18	11: mflo \$a2 * resultado de x*3
☐	0x00400018	0x00270001	mult \$18, \$16	12: mult \$a3, \$a0
☐	0x0040001c	0x0000a002	mflo \$20	13: mflo \$a4 * resultado de x*4
☐	0x00400020	0x00200001	mult \$20, \$16	14: mult \$a4, \$a0
☐	0x00400024	0x0000a002	mflo \$21	15: mflo \$a4 * resultado de x*5
☐	0x00400028	0x0010042a	srlt \$21, \$20	17: srt \$t1, \$a0, \$zero <= srt less than -> \$a0 for menor que zero, srt ti como 1, se nao, 0.
☐	0x0040002c	0x01200001	beq \$5, \$0, 0x0000000001	18: beq \$t1, \$zero, maior
☐	0x00400030	0x00100010	0x00000040	19: j memor
☐	0x00400034	0x022a0001	addsl \$10, \$19, 0x00000000...	21: addl \$t2, \$a3, 1
☐	0x00400038	0x0ad00004	sw \$10, 0x00000000(\$5)	23: sw \$t2, 4(\$t0)
☐	0x0040003c	0x00100012	0x000400048	24: j fim
☐	0x00400040	0x0030ffff	addsl \$21, \$20, 0xffff	25: addl \$t3, \$t4, 0xffffffffff

Data Segment

Address	Value (+0)	Value (+4)	Value (+8)	Value (+c)	Value (+10)	Value (+14)	Value (+18)	Value (+1c)
0x10010000	0x00000002	0x00000009	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x10010020	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x10010040	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x10010060	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x10010080	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x100100a0	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x100100c0	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x100100e0	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x10010100	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x10010120	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x10010140	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x10010160	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x10010180	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x100101a0	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000

Registers

Coproc 1

Coproc 0

Name	Number	Value
\$zero	0	0x00000000
\$at	1	0x00000000
\$v0	2	0x00000000
\$v1	3	0x00000000
\$a0	4	0x00000000
\$a1	5	0x00000000
\$a2	6	0x00000000
\$a3	7	0x00000000
\$a4	8	0x00100000
\$t0	9	0x00000000
\$t1	10	0x00000000
\$t2	12	0x00000000
\$t3	13	0x00000000
\$t4	14	0x00000000
\$t5	15	0x00000000
\$t6	16	0x00000000
\$t7	17	0x00000000
\$a0	18	0x00000002
\$a1	19	0x00000000
\$a2	20	0x00000004
\$a3	21	0x00000000
\$a4	22	0x00000010
\$a5	23	0x00000020
\$a6	24	0x00000000
\$a7	25	0x00000000
\$t8	26	0x00000000
\$t9	27	0x00000000
\$a8	28	0x00000000
\$t1	29	0x00000000
\$gp	30	0x00000000
\$ap	31	0x00000000
\$fp	32	0x00000000
\$ra	33	0x00000000
\$pc	34	0x00000000
\$t1	35	0x00000000
\$t0	36	0x00000000